Levers and limits of active control in cross-situational word learning

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Abstract

Learners who choose to hear words they do not yet know can recover much of the cost that a Zipfian vocabulary imposes on learning from random input. Analyses of this benefit, including our own, have idealized the learner's control: it chooses only the target word, on every episode, from the entire vocabulary, at no cost. We relax each idealization in simulations of an eliminative cross-situational learner. (1) Which lever: with small contexts, choosing the target does most of the work; with 100 referents per context, learning never finishes under Zipfian distractors unless the learner also controls what else is in view. Cross-word mutual exclusivity also rescues learning, but leaves target choice with nothing to do. (2) How much control, and over what: a learner active on a quarter of episodes gains most of the full benefit, but choice sets that over-represent frequent words can make learning up to 11 times slower, because words not yet encountered dilute the familiarity signal that context control relies on. (3) When: becoming active late in learning is slightly better than spreading the same autonomy evenly, while stopping early is much worse. (4) At what cost: on real image captions, staying with the current scene beats seeking a new one once switching scenes costs two or more episodes, although the difference is small. The benefit of active control depends on which levers a learner has and how they combine, not only on how much control the learner has.

Keywords: Word learning; Cross-situational learning; Active learning; Self-directed learning; Zipf's law; Development

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Introduction

Children hear words in a world that is far from uniform. Both the words caregivers say and the objects infants see follow highly skewed, Zipfian distributions (Clerkin, Hart, Rehg, Yu, & Smith, 2017; Lavi-Rotbain & Arnon, 2023), and under such distributions a learner who takes input as it comes must wait a long time for the rarest words. In a companion paper (Hidaka, Torii, & Kachergis, in preparation), we showed that this wait multiplies the number of learning episodes needed by roughly $\ln W$ for a vocabulary of W words, and that a learner who influences its input can recover much of the cost. A learner choosing among scenes can make rare words more likely, and an incremental learner who targets words it does not yet know learns 2–10 times faster than a passive one by the time it knows 50–90% of its vocabulary.

That analysis, like most formal treatments of active learning (Settles, 2012), idealized the learner's control in four ways. The learner chose only which word to hear, not what else was in view. It chose on every episode, from the whole vocabulary, always preferring the least-known word. Its control was the same from the first episode to the last. And obtaining a situation was free. None of these is true of children. Children are active learners (Foushee, Srinivasan, & Xu, 2023; Gureckis & Markant, 2012): they direct their gaze, hands, and requests toward what interests them (Partridge, McGovern, Yung, & Kidd, 2015; Yu, Smith, Shen, Pereira, & Smith, 2009), prefer stimuli of intermediate rather than minimal or maximal complexity (Kidd, Piantadosi, & Aslin, 2012), and adapt their information search to the environment from about age three (Ruggeri, 2022). But their control is partial. Much of what they hear is not directed by them at all. They choose among the objects at hand, not among every object they will ever name. Their capacity for control grows with motor development, which reshapes what infants see and do (Iverson, 2010; Kretch, Franchak, & Adolph, 2014; Walle & Campos, 2014). And, like adults, they do not always choose well (Bazhydai, Wong, Altmann, Jones, & Westermann, 2026; Tauber, Dunlosky, Rawson, Wahlheim, & Jacoby, 2013).

This paper asks how the benefit of active control depends on these features. We organize the question around four aspects of control.

Which lever. Caregivers and children shape not only what is named but what else is in view when it is named: in play, they often arrange for a single object to dominate the child's view (Suanda, Barnhart, Smith, & Yu, 2018; Yu et al., 2009), and children exclude already-known objects as candidate referents for a new word (Grassmann, Schulze, & Tomasello, 2015). Study 1 asks how control over the target word and control over the accompanying distractors combine, and how this depends on the number of referents in view.

How much, and over what. Study 2 asks how much of the benefit survives when the learner is active only some of the time, when it chooses only among a small set of available words, and when it prefers words of intermediate familiarity, as the Goldilocks effect suggests infants do (Kidd et al., 2012).

When. Infants begin with little control over what they experience and gain it as they learn to reach, crawl, and walk. Because active selection is worth more late in learning, when only rare words remain, the timing of control could matter in its own right. Study 3 compares developmental trajectories from passive to active, and the reverse, with constant levels of control.

At what cost. Gaining a new situation takes effort; asking more about the current one may take less. This is the patch-leaving problem of foraging theory (Charnov, 1976), which has been applied to search in memory and other cognitive domains (Hills, Todd, Lazer, Redish, & Couzin, 2015). Study 4 asks, using real image captions, when a learner should stay with the current scene and when it should move on.

Across the four studies, the same theme emerges: active control helps by directing the learner's episodes toward its current bottleneck, and different levers act on different bottlenecks. Levers that act on different bottlenecks need not combine, and constraints that change where the bottleneck lies can change which lever matters.

General method

All studies use the eliminative cross-situational learner of the companion paper (after Siskind, 1996; Vogt, 2012); Study 1 also uses its guess-test learner. There are W words, each paired with one of W referents. In each episode the learner hears one word, the *target*, in a situation containing its referent and $C - 1$ distractor referents. The eliminative learner keeps, for each word, the set of referents present every time that word has been heard, and learns the word once a single candidate remains. The guess-test learner keeps one guess per word, proposing a referent from the current situation that no other word claims, discarding the guess if the word is later heard without it, and learning the word once its guess is correct.

Word and referent frequencies are Zipfian, $p_r \propto r^{-a}$ over frequency rank r, with $a = 1$ unless stated ($a = 0$ is uniform). Two choices define each episode:

- *Target selection.* A *passive* learner's target is drawn by frequency. An *active* learner's target is drawn among words not yet known.
- *Context selection.* Under *random* context selection, the $C - 1$ distractors are drawn by frequency. Under *familiar* context selection, each referent's frequency is divided by one plus the number of words whose candidate sets still contain it, so that distractors tend to be referents the learner has already ruled out for most words. This formalizes the use of known objects to disambiguate new words (Grassmann et al., 2015), and is consistent with evidence that discourse and attention, not only visual presence, determine which referents are candidates (Frank, Goodman, & Tenenbaum, 2009; Frank, Tenenbaum, & Fernald, 2013).

Unless stated, vocabularies have $W = 1{,}000$ words, and we record the number of episodes until 50%, 90%, and 99% of the vocabulary is learned, with 30 replications per condition. Simulations at $C = 100$ and at $W \geq 10{,}000$ used the companion paper's fast reimplementation of the eliminative learner, which agrees with the reference implementation within sampling error on every condition both can complete. Code and results are available in the project repository (`corpusXSL/analysis`).

Study 1: Which lever? Target selection vs. context selection

The companion paper varies one lever: which word is the target of an episode. But a learner or caregiver also influences what else is in view when a word is heard. This study crosses target selection (passive or active) with context selection (random or familiar), and shows that at larger context sizes the second lever, not the first, decides whether elimination-based learning can finish at all.

We simulated the eliminative learner at $C = 10$ and $C = 100$, on either side of Vogt (2012)'s reported tractability boundary of $C/W \leq 0.1$ (his C/M), for uniform and Zipfian vocabularies of $W = 1{,}000$ words, and recorded episodes to learn 99% of the vocabulary. Runs were capped at 5×10^8 episodes, and capped runs are reported as such.

Results: context size decides which lever matters

Table 1 and Figure 1 summarize the results at $a = 1$. At $C = 10$ ($C/W = 0.01$), active target selection does most of the work (19.1x), and familiar-context selection adds only a modest further benefit (2.0x alone, 28.6x combined), because elimination is already fairly effective when the distractor set is small.

At $C = 100$ ($C/W = 0.1$), random-context learning does not merely slow down under a Zipfian distribution: it fails. No passive or active run with random distractors reached even half the vocabulary within 5×10^8 episodes. The reason is the eliminative learner's requirement that every competing referent be absent from at least one of a word's contexts. With 99 frequency-weighted distractors per context, the most frequent referent ($p \approx 0.13$ at $a = 1$) is absent from any given context with probability only about 5×10^{-10} (computed exactly[1]), so every other word would need on the order of

[1] By the exponential-clock representation of weighted sampling without replacement: each referent j receives an independent clock $E_j \sim \text{Exp}(p_j)$, the context's distractors are the $C - 1$ referents whose clocks ring first, and referent r is absent iff at least $C - 1$ other clocks ring before E_r. The resulting one-dimensional integral over a Poisson-binomial tail agrees with direct simulation wherever the probability is large enough to simulate; see `corpusXSL/analysis/c100_absence_probability.R`.

2×10^9 exposures merely to rule out that one competitor. Active target selection cannot help, since it changes which word is the target but not which distractors accompany it. Familiar-context selection, which does change the distractors, restores learning (446,962 episodes), and once distractors are controlled, active target selection adds a further 28-fold advantage (15,819 episodes). At $C = 100$, controlling what else is in view is a precondition for learning at all.

This failure is a matter of degree of skew, not an all-or-none property of $C = 100$. Sweeping the Zipf exponent at $C = 100$ (Figure 2, right) shows passive random-context learning slowing steeply but continuously, from 9,904 episodes at $a = 0$ to 40,456 at $a = 0.5$, 1.45 million at $a = 0.7$, 47 million at $a = 0.8$, and beyond our 10^8-episode cap from $a = 0.85$ on, tracking the exact probability that the most frequent referent is absent from a given context (0.17 at $a = 0.5$, 2.6×10^{-4} at $a = 0.8$, 1.5×10^{-6} at $a = 0.9$). Active target selection helps throughout this range (e.g., 12x at $a = 0.8$) but succumbs in turn (38 million episodes at $a = 0.85$, beyond the cap at $a = 0.9$). Only familiar-context selection keeps learning fast as skew grows (at $a = 0.9$: 106,039 passive and 10,646 active episodes).

Context size, not the ratio C/W

The same failure appears at larger vocabularies (Table 2). $C = 100$ at $W = 10{,}000$ has the same ratio, $C/W = 0.01$, as $C = 10$ at $W = 1{,}000$, where learning is easy, and $C = 100$ at $W = 60{,}000$ has a smaller ratio still (0.0017). Yet at both sizes, with random distractors, no run, passive or active, learned even 10% of the vocabulary within 10^8 episodes (10 replications per condition). With 99 distractors per context, the most frequent referent ($p \approx 0.10$ at $W = 10{,}000$ and 0.086 at $W = 60{,}000$) is absent from a given context with probability only about 1.5×10^{-6} and 2.7×10^{-5} respectively, so the rarest words would need on the order of 10^{10} episodes or more merely to rule out that one competitor. Familiar-context selection again restores learning. Under Zipfian distractors, then, whether elimination is feasible depends on the absolute context size together with the skew of the referent distribution, rather than on the ratio C/W alone.

C/W does, however, shape how large the active advantage is once learning is possible. With familiar context at $C = 100$, the active advantage is 28-fold at $W = 1{,}000$ but only 7.6-fold at 10,000 and 9.3-fold at 60,000. Recording how many exposures each word needed before it was learned (10 replications per condition) shows why. Target selection does not change how many exposures a given word needs: passive and active learners need the same number, frequency band by frequency band. What differs is how each learner pays for them. An active learner never spends an episode on a word it already knows, so its total is essentially the sum of all words' needs, and under familiar context at $C = 100$ that sum is dominated by the most frequent words, whose competitors are other frequent referents: the top 1% of words need a median of 109–229 exposures each, the bottom half only 5–7. A passive learner instead waits for its last few rare words to be sampled often enough, so each exposure a rare word needs costs it roughly the inverse of that word's sampling probability. At $W = 1{,}000$ these last words are unusually hard–the last 1% learned need a median of 34 exposures, against 5–6 at $W = 10{,}000$ and 60,000–which inflates the passive learner's time, and hence the speedup. Varying C at fixed M indicates that this hard tail grows with the share of all referents present in each context (5 replications per condition): at $W = 1{,}000$, the last 1% of words need a median of 3, 6, and 34 exposures at $C/W = 0.01$, 0.03, and 0.1, and at $W = 10{,}000$, 6, 10, and 297 at the same ratios. When each context contains a tenth of all referents, a competitor that survives one exposure has a good chance of reappearing in the next.

A learner that does not need full elimination

The guess-test learner of the companion paper, which keeps a single guess per word and counts a word as learned once its guess lands on the true referent, never needs every competing referent to be absent from some context. Accordingly it still learns at $C = 100$, $a = 1$ with random distractors (326,494 episodes passive, 176,578 active), where the eliminative learner never finishes, and active target and familiar-context selection give it only modest speedups (1.8x and 3.8x). Combining active target

selection with familiar-context selection produced a bimodal outcome for this learner: four of five diagnostic replications finished in 135,000–212,000 episodes, but one exceeded a three-million-episode budget, apparently because familiarity-weighted distractor selection can concentrate the distractor pool onto a few already-claimed referents, leaving the last hard-to-place words without an unclaimed candidate to propose.

Cross-word mutual exclusivity

Every learner above narrows a word's candidates independently. Reisenauer, Smith, Smith, and Blythe (2013) show that a cross-word mutual exclusivity constraint–confirming one word's meaning immediately excludes that referent for every other still-unlearned word–can rescue an elimination-based learner from exactly the slowdown seen at large C. We added this rule to the eliminative learner, with cascades of further learning events allowed within an episode, and compared it against the independent model at $W = 1{,}000$, $a = 1$, random context (Table 3; 30 replications per cell at $C = 10$, and 200 at $C = 100$, where learning time is heavy-tailed; no replication was censored).

At $C = 100$, mutual exclusivity delivers the predicted rescue: both passive and active learners now reach 99% of the vocabulary, in well under a million episodes on average. The trajectory is unusual, however. Essentially the whole vocabulary is learned in a single cascade: in 396 of 400 replications, the 50%, 90%, and 99% checkpoints were all reached within 10 episodes of one another, after a long initial period in which almost nothing was learned. And active target selection confers little benefit: 686,349 versus 771,125 episodes (medians 441,212 and 572,126; $z \approx 1.3$).

Both features have the same explanation. Nothing can cascade until at least one word has been learned by ordinary elimination, which requires ruling out every competitor for that word–least severe for the most frequent word, whose own referent is the competitor otherwise hardest to rule out. Once that first word is learned, its referent is removed from every other word's candidate set, which unblocks the next, and

so on. The wait for the first learning event dominates the total time, and during that wait active target selection is inert: its advantage comes from never spending an episode on an already-known word, and while no word is yet known, that restriction does nothing. Holding the random seed fixed, active and passive target selection produce identical target sequences until the first word becomes known. Mutual exclusivity and active target selection, both beneficial elsewhere, therefore do not compound at large C–a second case, alongside guess-test's bimodality, of levers that fail to compose.

Study 2: How much autonomy, and over what?

In the companion paper and in Study 1, an active learner is idealized in two ways: it exercises choice on every episode, and whenever it does, it can choose among the entire vocabulary, preferring any word it does not yet know. Real learners exercise control only some of the time; they choose among whatever happens to be available (the toys in the room, not every object they will ever name); and they may not simply prefer the least-known item, since infants preferentially attend to stimuli of intermediate familiarity (the "Goldilocks effect"; Kidd et al., 2012). This study relaxes each idealization in turn.

Setup

We extended the eliminative learner's target selection in three ways. *Partial autonomy*: each episode is active with probability active_prob $\in \{0.25, 0.5, 0.75\}$ and passive otherwise, a constant blend of active and passive selection. *Goldilocks targeting*: among not-yet-known words, the learner prefers those it has encountered about twice, weighting each by $\exp(-(n-2)^2/8)$ times its frequency, where n is the number of times the word has been the target so far. *Bounded choice*: on each active episode the learner may choose only among a choice set of $k \in \{100, 50, 20, 5\}$ words, drawn without replacement in proportion to frequency (so that commonly-encountered things are more likely to be at hand), preferring an unknown word within the set and making a passive draw from the set if it contains none.

We ran each condition, together with pure passive and fully active selection, in four settings for a Zipfian vocabulary of $W = 1{,}000$ words: $C = 10$ with random and with familiar context selection ($a = 1$), and two settings at $C = 100$. Study 1 shows that at $C = 100$, $a = 1$, the eliminative learner never completes with randomly-composed contexts, even under fully active selection, so there we use the two $C = 100$ settings in which learning does complete: familiar context at $a = 1$, and random context at $a = 0.75$, the steepest skew at which random-context learning finished in every run. All conditions were simulated with the companion paper's fast reimplementation of the eliminative learner, which we validated for each of these target-selection rules against the reference implementation (200 replications per condition at $C = 10$, 30 at $C = 100$).

Results

Table 4 and Figure 3 report each condition's episodes to learn 99% of the vocabulary relative to fully active selection in the same setting.

Partial autonomy shows steeply diminishing returns. In every setting, a learner who is active on only a quarter of episodes already captures most of active selection's benefit: measured on a log scale, between two-thirds and four-fifths of the full passive-to-active speedup (e.g., at $C = 10$ with random context, 7.1x of a possible 19.0x). Each further increment of autonomy buys less. Study 3 asks whether, beyond its amount, the timing of autonomy matters.

Goldilocks targeting matters in only one setting. Preferring words encountered about twice makes no difference in three of the four settings (0.996–1.04x), but nearly halves learning time at $C = 100$ with familiar context (0.56x).

The cost of bounded choice depends on the setting. In three settings, restricting choice costs little until the choice set is small, and costs more the smaller it is (1.03–1.09x with 100 words to choose from, rising to 2.8–4.1x with 5). At $C = 100$ with familiar context, the pattern is different: even a choice set of 100 makes learning 5.5 times slower, and the cost is not monotone in the size of the set (11.3x with 20 words, 10.4x with 5).

A diagnostic isolates the cause of this last pattern. Drawing the choice set uniformly rather than in proportion to frequency, and otherwise leaving everything unchanged, removes the cost almost entirely at $C = 100$ with familiar context: 15,124 episodes with a uniform set of 100 words and 15,894 with 20, against 15,876 for unrestricted choice (30 replications each). The cost therefore comes not from restricting choice as such but from choice sets that over-represent frequent words: when the set is drawn by frequency and the target is then chosen from it by frequency again, a rare word's chance of being targeted scales roughly with the square of its frequency, and rare words are rarely targeted at all. Together with the Goldilocks result, this suggests that at $C = 100$ with familiar context, learning is fastest when exposures are spread across the whole vocabulary, including its rare words.

Mechanism: unseen words dilute the familiarity signal

Why should spreading exposures matter, and only with familiar context at large C? Familiar-context selection weights each candidate distractor r by $1/(n_r + 1)$, where n_r is the number of words that still hold r as a possible meaning. A word that has never been the target of an episode has eliminated nothing, so it still holds every referent as a possible meaning, and each such unseen word adds the same amount to every referent's n_r. While many words are unseen, this shared count swamps the differences between referents, and familiar-context selection behaves almost like random selection. On this account, familiar context only starts to help once most words have been encountered at least once, and policies that concentrate targets on frequent words delay that point.

We tested this account in two ways, both at $C = 100$, $a = 1$ with familiar context (Table 5, Figure 4). First, we removed the dilution by computing each referent's familiarity from encountered words only, i.e. discounting the contribution of unseen words. If dilution is the mechanism, this should largely remove the differences between policies, and it does. The frequency-weighted choice set of 100 words goes from 5.7 times slower than unrestricted choice to 1.3 times; Goldilocks targeting goes from 0.57 to 0.93. At the 50% checkpoint, the choice-set cost disappears entirely (5,547 episodes

vs. 5,127 unrestricted, down from 84,671). What remains for small frequency-weighted choice sets (3.3x for 20 words) arises late in learning, and reflects the separate, direct problem that rare words are seldom in the choice set and so are seldom targeted. Removing the dilution also halves learning time for unrestricted active selection (15,770 to 7,803 episodes), while leaving passive sampling essentially unchanged (466,339 to 451,567), whose bottleneck is instead how rarely its rare words are targeted.

Second, we tracked each policy's trajectory under the original weighting. The frequency-weighted choice set leaves many words unencountered for a long time–634 after 10,000 episodes, against 63 for unrestricted choice and 34 for a uniform choice set–and throughout this period, 96–98% of a target's remaining competing referents reappear among its distractors, escaping elimination, against 66–84% for the other policies. Goldilocks targeting, which favors words encountered only once or twice, brings the number of unencountered words down fastest (33 after 5,000 episodes) and has the lowest competitor survival (about 50%).

Why only at large C? At $C = 10$ elimination is fast even with random distractors, so familiar-context selection adds relatively little (Study 1 finds a 2-fold benefit at $C = 10$, against a difference between learning and not learning at all at $C = 100$), and a policy that delays its benefit loses correspondingly little.

These tests also suggest a modeling point. The dilution arises from treating a never-encountered word as "still considering every referent", which is the natural bookkeeping for an eliminative learner, but not the only plausible one: a learner that simply ignores words it has not yet heard when judging how familiar an object is learns twice as fast under unrestricted active selection, and is far less sensitive to how its choice set is composed.

Scope

Two conclusions seem robust across the settings we tested: partial autonomy has steeply diminishing returns, and most of active selection's benefit survives a learner who exercises it only occasionally. The results for Goldilocks targeting and bounded choice

are more conditional. Their size, and at $C = 100$ with familiar context their direction relative to other settings, depends on context size and on whether distractors are controlled, and at least for bounded choice, on what the available choice set contains rather than merely how large it is. If children's accessible objects are, like the objects they encounter generally, dominated by common ones, the composition of what is at hand may matter as much as the freedom to choose among it. We have tested only the eliminative mechanism, four settings, and frequency-weighted or uniform choice sets; how real choice sets are composed, and how large they are, are empirical questions we do not address.

Study 3: A developmental passive-to-active trajectory

Studies 1 and 2 hold the degree of autonomy fixed for the whole simulated lifetime of the learner. Real infants do not work this way: a newborn is carried from room to room with essentially no say over which scenes it experiences, and only gradually acquires the visual, postural, and locomotor control–head-turning, pointing, reaching, eventually crawling and walking–that lets it actively select what to attend to. This study asks whether that developmental trajectory itself, quite apart from its eventual endpoint, has consequences for how much of active selection's benefit a learner realizes.

Motivation and setup

The companion paper's completion-checkpoint analysis already implies an answer is possible: the active/passive gap widens over the course of learning (roughly 4-fold at 50% completion, 9-fold at 90%, 19-fold at 99%, for the eliminative learner at $C = 10$), meaning active selection's marginal value is low early–common words get learned quickly under either policy–and high late, when only rare stragglers remain. If a fixed "active budget" is worth more late than early, a learner who starts passive and becomes active over time should outperform a constant blend that spends the same budget evenly, and should do considerably better than a learner who front-loads the same budget into the early, low-value period.

We extended the eliminative learner with an optional `active_prob_fn`: a function of the episode count, re-evaluated every episode, in place of a constant `active_prob`. We compare a forward, linear ramp from passive to active, active_prob$(t) = \min(1, t/T)$, against its mirror image, a reverse ramp from active to passive, active_prob$(t) = \max(0, 1 - t/T)$. Over their first T episodes the two ramps are active equally often, but the reverse ramp places that activity where the checkpoint analysis says it is worth least. Both are indexed by episode count rather than fraction of vocabulary learned, since real motor and attentional development runs on its own maturational clock and is not contingent on how much vocabulary a particular child happens to have acquired so far; this makes the ramp's timescale T a free parameter, swept here (at $W = 1{,}000$, $C = 10$, $a = 1$, 30 replications per condition) rather than fixed at a single guessed value.

Results

Table 6 gives the reference points–pure passive and active selection, and constant blends at 13 intermediate levels of autonomy–and Figure 5 shows both ramp directions across seven timescales, $T \in \{1{,}000, 2{,}000, 3{,}000, 5{,}000, 8{,}000, 12{,}000, 20{,}000\}$ episodes.

Forward ramps are slower the longer the learner stays passive, rising smoothly from 4,875 episodes at $T = 1{,}000$, within 5% of pure active selection, to 10,573 at $T = 20{,}000$. Reverse ramps behave very differently. At $T = 1{,}000$ the reverse ramp takes 87,550 episodes, close to pure passive selection (89,857), and it stays near the passive level through $T = 8{,}000$ (57,248) before falling sharply by $T = 12{,}000$ (6,049). The threshold lies where the ramp's midpoint, $T/2$, is close to the time a fully active learner needs to finish (4,681 episodes). When T is large, a learner who starts active finishes before its autonomy has decayed far. When T is small, autonomy decays long before the vocabulary is complete, and the long remainder of the run is passive.

The contrast between directions at a given T confounds timing with the total amount of autonomy, because a learner who becomes active stays active for the rest of its run, while one who becomes passive stays passive. At $T = 1{,}000$, the forward learner

is active on about 90% of its episodes and the reverse learner on fewer than 1%. To isolate timing, we compared each ramp with a constant blend of the same autonomy. For each run we computed its realized autonomy–the mean proportion of active episodes over the episodes it actually ran, which is exact because the ramp depends only on the episode count–and compared each ramp's mean episodes with the constant-blend curve of Table 6 at the ramp's mean realized autonomy, interpolating on a log scale. Forward ramps were 4–13% faster than constant blends of matched autonomy, and the gain grew with the timescale: 4% at $T = 1{,}000$ (autonomy 0.90), 11% at $T = 8{,}000$ (0.46), and 13% at $T = 20{,}000$ (0.26). Reverse ramps that lose control before the vocabulary is learned ($T \leq 8{,}000$) were 1.2 to 2.3 times slower than matched blends, the largest cost at $T = 5{,}000$–$8{,}000$ (autonomy 0.03–0.07); those that finish while still mostly active ($T \geq 12{,}000$) were within 4% of them. Timing therefore matters in the direction the checkpoint analysis predicts–autonomy is worth more late in learning than early–but for a learner who becomes active, its effect is modest compared with the effect of how much of the run is spent under control.

Scope

These results support a narrow claim. A developmental trajectory from little control to more control loses little of active selection's benefit if control arrives quickly relative to the time needed to learn the vocabulary, and it is slightly more efficient than the same amount of control spread evenly. Losing control early is much more costly. We make no claim about the timescale of real infant development, and we have tested only linear ramps, indexed by episodes rather than vocabulary or age, for one learning mechanism, context size, and Zipf exponent. Estimating how quickly children's self-directed exploration increases, for example from egocentric video recorded across the transitions to crawling and walking (Kretch et al., 2014; Sullivan, Mei, Perfors, Wojcik, & Frank, 2021), is a natural next step.

Study 4: Staying or moving on: scene-switching cost on real image captions

The companion paper tests target selection on real, naturalistic co-occurrence structure using Flickr8k (Hodosh, Young, & Hockenmaier, 2013): 8,108 photographs, each independently described by five human-written captions, but its learner always moves on to a new photo after each caption. Real captions let us ask a further question the synthetic simulations cannot pose at all: beyond choosing *which* word to target, a learner who can request another caption of the scene already in view, rather than only ever moving on to a new one, faces a second active choice with no synthetic analogue–and it is not obvious in advance which side of that choice should win.

Setup

We treat each caption as a situation: a target word plus whichever other content words co-occur in that sentence, playing the role of distractors. After excluding function words (articles, prepositions, pronouns, and similar) via a hand-curated stoplist and words occurring only once, 4,318 content words remain, averaging 5.76 per caption–a naturally-occurring analogue of context size C, close to the companion paper's own $C = 10$ condition. A word is eliminative-learnable if the intersection of the word-sets of every caption in which it appears narrows to the word itself; 83.3% (3,598 words) are learnable this way, and the remainder are stuck with at most 8 residual confusable alternatives, not a pathological blowup.

We compare three policies. *Passive*: sample a photo uniformly at random, reveal one of its captions at random, and always move on. *Active, move-on*: target the rarest not-yet-learned learnable word, sample a photo containing it, reveal one caption, and always move on. *Active, linger*: as move-on, but remain on the current photo–revealing another of its remaining captions–for as long as doing so is still shrinking some unknown word's candidate set, moving on only once the current photo stops paying off. A `move_cost` parameter charges additional episodes for switching to a new photo, formalizing the idea that eliciting another description of the scene already in view may be cheaper than obtaining a new scene built around a specific word; `move_cost`$= 0$

makes the two active policies differ only in information, not economy.

Results

Table 7 and Figure 6 report mean episodes to reach 50%, 90%, and 99% of the learnable vocabulary (30 replications per cell). The companion paper reports the passive vs. active move-on comparison at zero switching cost as its real-data test of target selection (23-fold at 99%); here we focus on the cost of switching. Even at the highest scene-switching cost we tested, both active policies still finish 99% of the vocabulary in under half the episodes passive sampling requires.

The linger-vs-move-on choice itself shows a genuine crossover rather than one policy dominating throughout. At `move_cost` 0 and 1, always moving to a fresh photo wins (by 13% and 3% at 99%): a new scene supplies more diverse elimination context than a second, independently-written description of the same one, so the extra information from a new scene outweighs the (zero or small) cost of obtaining it. Between `move_cost` 1 and 2, the ranking flips, and lingering's advantage widens as switching gets more expensive: lingering is 1% faster than moving on at `move_cost`=2, 3% at 3, 5% at 5, and 7% at 10. In other words, whether "ask for more about what's already in view" is the better strategy is not fixed; it depends on the relative cost of obtaining a new situation versus refining the current one, exactly the kind of parameter the companion paper's synthetic simulations, which never separated situation-content from situation-acquisition cost, could not have exposed.

This result should be read as a demonstration that the mechanism generalizes to real co-occurrence structure and that the linger-versus-move-on tradeoff is a real, locatable phenomenon, not as a claim about what value `move_cost` actually takes for a child or caregiver in practice–we have no direct estimate of that value, and supplying one is a natural target for future work. It is also worth being explicit that Flickr8k captions are adult-crowdsourced descriptions of generic photographs, not child-directed speech or a child's own visual experience, so this study speaks to whether the mechanisms in the companion paper behave sensibly on naturalistic language statistics,

not to developmental realism specifically.

General discussion

The companion paper established that active control over learning situations can recover much of the cost of a Zipfian vocabulary. The four studies here ask what happens to that benefit when control is closer to what children have. The answers depend less on how much control a learner has than on which lever it acts on and where the learner's bottleneck lies.

Levers act on bottlenecks

An eliminative learner can be held up in two ways. It can wait for rare words to be heard often enough, or it can wait for frequent competitors to be absent from a word's contexts. Target selection removes the first bottleneck, since an active learner never spends an episode on a word it already knows. Context selection removes the second, since it fills contexts with referents already ruled out for most words. Which bottleneck dominates depends on the number of referents in view and on the skew of their distribution (Study 1). With 10 referents in view, competitors are easily ruled out, and target selection does most of the work. With 100, the most frequent referents are almost never absent from a context, and learning never finishes unless distractors are controlled–whether by familiar-context selection or by cross-word mutual exclusivity. Once distractors are controlled, target selection again adds a large benefit.

This view also explains when levers fail to combine. Mutual exclusivity removes the competitor bottleneck by a cascade that begins with the first word learned; before that word is learned, target selection has nothing to act on, because no word is yet known, and it adds almost nothing (Study 1). Frequency-weighted choice sets restore the rare-word bottleneck that target selection removed, and they also delay familiar-context selection, because familiarity is computed over words, and words not yet heard dilute the signal (Study 2). For the guess-test learner, combining active target and familiar-context selection occasionally left the last few words without any

unclaimed referent to propose (Study 1). In each case, two forms of control that help separately interfere because one changes the state that the other depends on.

Partial, constrained, and developing control

The most robust result is that partial control goes a long way. A learner active on a quarter of episodes gains between two-thirds and four-fifths of the full benefit, on a log scale, in every setting tested (Study 2). This is encouraging for the developmental relevance of active learning, because children's control over their input is intermittent: much of what they hear is overheard, initiated by others, or about things they did not choose.

What a child chooses among may matter more than how often they choose. Restricting choice to small sets of words costs little when the sets are drawn uniformly from the vocabulary, but a great deal when they over-represent frequent words, as the objects within a child's reach plausibly do (Clerkin et al., 2017). Preferring words of intermediate familiarity (Kidd et al., 2012) made no difference in most settings; where it helped, the gain came from spreading early exposures across the vocabulary, not from intermediate familiarity as such (Study 2).

Timing matters, in the direction expected from the growing value of active selection over learning, but modestly (Study 3). A learner who becomes active over time does 4–13% better than one with the same autonomy spread evenly, and a learner who starts active and loses control before the vocabulary is learned does up to 2.3 times worse. The largest effect of a developmental trajectory is through how much of the learner's run is spent under control, which depends on how quickly control arrives relative to how quickly the vocabulary is learned.

The cost of new situations

On real captions, whether to stay with the current scene or seek a new one depends on the relative cost of the two (Study 4): when switching scenes is free or cheap, a new scene is worth more than another description of the same one, and once it costs two or more episodes, staying is better. The difference between policies was small,

at most 7% at the costs tested, so this study shows that the trade-off exists and where it reverses, more than that it matters a great deal. Like foraging animals deciding when to leave a patch (Charnov, 1976), a learner should stay with a scene while it still yields information at a higher rate than a new scene would, after the cost of finding one.

Limitations and future directions

Most of our results come from a single learning mechanism, the eliminative learner, and Studies 2 and 3 use a single vocabulary size and Zipf exponent. The guess-test learner behaved differently in Study 1, and extending Studies 2–4 to it is an obvious next step. Our policies remain idealized: the active learner knows exactly which words it has learned, the ramp in Study 3 is linear and indexed by episodes rather than by age, and the costs in Study 4 are free parameters. Flickr8k captions are adult descriptions of photographs, not child-directed speech or a child's view. Each of these idealizations could be informed by data. Egocentric recordings of infants' everyday experience (Clerkin et al., 2017; Sullivan et al., 2021) could be used to estimate how often children's attention is self-directed, which objects are available to choose among, how quickly self-directed exploration increases with motor development (Kretch et al., 2014), and how often a caregiver names a second object in the same scene rather than a new scene being sought. With such estimates, the present framework could predict not just whether active control helps, but how much each lever contributes over the course of a particular child's vocabulary growth.

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Table 1

Mean episodes to learn 99% of a Zipfian ($a = 1$), $W = 1000$-word vocabulary, eliminative learner, 30 replications per cell. Speedup is relative to Passive/Random at the same C. At $C = 100$, no random-context run reached even 50% of the vocabulary within the 5×10^8-episode cap, so speedups there are lower bounds.

	$C = 10$ ($C/W = 0.01$)		$C = 100$ ($C/W = 0.1$)	
Condition	Episodes	Speedup	Episodes	Speedup
Passive / Random	89,454	1x	$> 5 \times 10^8$	1x
Active / Random	4,695	19.1x	$> 5 \times 10^8$	–
Passive / Familiar	44,607	2.0x	446,962	>1,100x
Active / Familiar	3,127	28.6x	15,819	>31,000x

Table 2

Mean episodes to learn 99% of a W-word vocabulary at $C = 100$, eliminative learner. ">": no run reached even 10% (at $W = 1{,}000$, even 50%) of the vocabulary within the stated cap.

Condition	$W = 1{,}000$	$W = 10{,}000$	$W = 60{,}000$
Zipfian ($a = 1$), Passive / Random	$> 5 \times 10^8$	$> 10^8$	$> 10^8$
Zipfian ($a = 1$), Active / Random	$> 5 \times 10^8$	$> 10^8$	$> 10^8$
Zipfian ($a = 1$), Passive / Familiar	446,962	866,781	5,453,160
Zipfian ($a = 1$), Active / Familiar	15,819	113,584	589,504
Uniform, Passive / Random	9,904	80,048	424,968
Uniform, Active / Random	3,718	26,261	128,363

Table 3

Mean episodes to learn 99% of a Zipfian ($a = 1$), $W = 1{,}000$-word vocabulary, eliminative learner, random context, with vs. without cross-word mutual exclusivity. Without mutual exclusivity, no $C = 100$ run reached even 50% of the vocabulary within 5×10^8 episodes, so those speedups are lower bounds.

Condition	Independent	Mutual exclusivity	Speedup
$C = 10$, Passive	89,454	23,184	3.9x
$C = 10$, Active	4,695	1,569	3.0x
$C = 100$, Passive	$> 5 \times 10^8$	771,125	>640x
$C = 100$, Active	$> 5 \times 10^8$	686,349	>720x

Table 4

Episodes to learn 99% of a $W = 1{,}000$-word Zipfian vocabulary, eliminative learner, relative to fully active selection in the same setting (first row: fully active episodes). Values above 1 are slower than fully active.

	$C = 10$, $a = 1$		$C = 100$, $a = 0.75$	$C = 100$, $a = 1$
Condition	Random	Familiar	Random	Familiar
Fully active (episodes)	4,692	3,132	628,691	16,060
Passive	19.0x	14.2x	10.6x	26.6x
25% active	2.67x	2.51x	2.33x	1.81x
50% active	1.67x	1.63x	1.58x	1.34x
75% active	1.24x	1.23x	1.21x	1.11x
Goldilocks	1.00x	1.04x	1.00x	0.56x
Choice set of 100	1.03x	1.09x	1.03x	5.51x
Choice set of 50	1.13x	1.15x	1.09x	8.01x
Choice set of 20	1.54x	1.43x	1.32x	11.3x
Choice set of 5	4.15x	3.32x	2.78x	10.4x

Table 5

Episodes to learn 99% of a $W = 1{,}000$-word Zipfian ($a = 1$) vocabulary at $C = 100$ with familiar context, relative to unrestricted active selection under the same familiarity measure (30 replications per cell). "All words": the original familiarity measure, in which every unencountered word counts as still holding every referent; "encountered words only": that contribution removed.

	Familiarity computed from	
Condition	All words	Encountered words only
Unrestricted active (episodes)	15,770	7,803
Choice set of 100, frequency-weighted	5.66x	1.34x
Choice set of 20, frequency-weighted	11.2x	3.32x
Choice set of 100, uniform	0.96x	1.01x
Goldilocks	0.57x	0.93x
Passive	29.6x	57.9x

Table 6

Mean episodes to learn 99% of a Zipfian ($a = 1$), $W = 1{,}000$-word vocabulary, eliminative learner, $C = 10$, for constant levels of autonomy (proportion of episodes on which the target is actively selected); 30 replications each. Study 2 reports some of the same blends, relative to fully active selection, from a separate set of simulations.

Proportion active	Episodes	Proportion active	Episodes
0 (passive)	89,857	0.30	11,048
0.02	41,853	0.40	9,190
0.05	29,433	0.50	7,803
0.075	24,317	0.60	6,869
0.10	21,116	0.75	5,829
0.15	16,871	0.90	5,073
0.20	14,332	1 (active)	4,681
0.25	12,633		

Table 7

Mean episodes to reach 50/90/99% of the 3,598-word learnable vocabulary, Flickr8k captions, eliminative mechanism. `move_cost` *is the number of extra episodes charged for switching to a new photo (passive sampling has no such switching decision).*

Condition	move_cost	50%	90%	99%
Passive	–	10,386	60,719	151,671
Active, move-on	0	2,779	5,643	6,528
Active, linger	0	3,125	6,511	7,515
Active, move-on	1	5,558	11,285	13,057
Active, linger	1	5,483	11,587	13,461
Active, move-on	2	8,338	16,928	19,585
Active, linger	2	7,841	16,664	19,408
Active, move-on	3	11,117	22,570	26,113
Active, linger	3	10,199	21,740	25,354
Active, move-on	5	16,675	33,856	39,170
Active, linger	5	14,915	31,893	37,247
Active, move-on	10	30,571	62,069	71,811
Active, linger	10	26,705	57,275	66,980

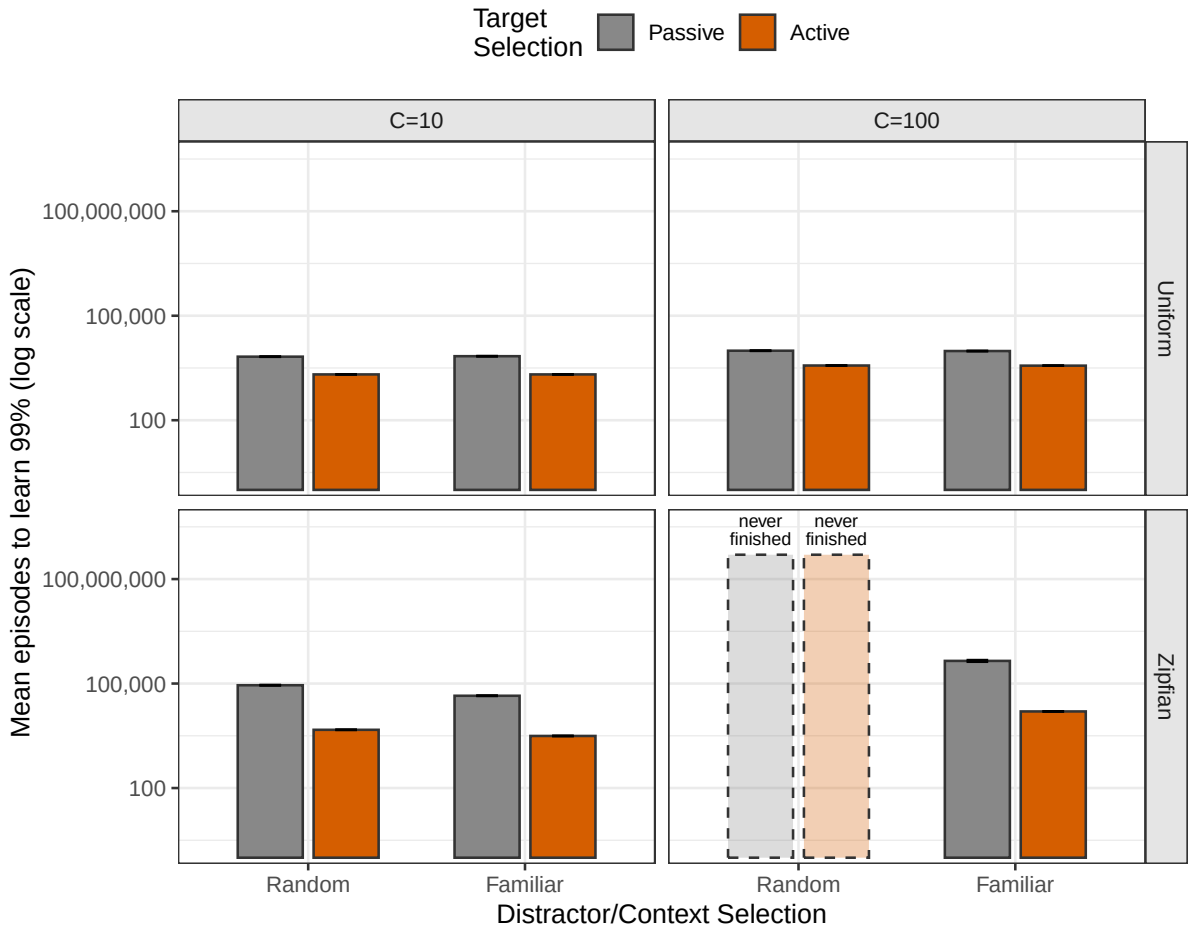


Figure 1. Mean episodes (log scale, ± 1 SE) for the eliminative learner to acquire 99% of a 1000-word vocabulary, by target selection (bar fill) and context selection (x-axis), for uniform and Zipfian ($a = 1$) distributions and $C = 10$ and $C = 100$. Faded, dashed bars mark conditions in which no run reached 99% (or even 50%) of the vocabulary within the 5×10^8-episode cap; they are drawn at the cap and are lower bounds.

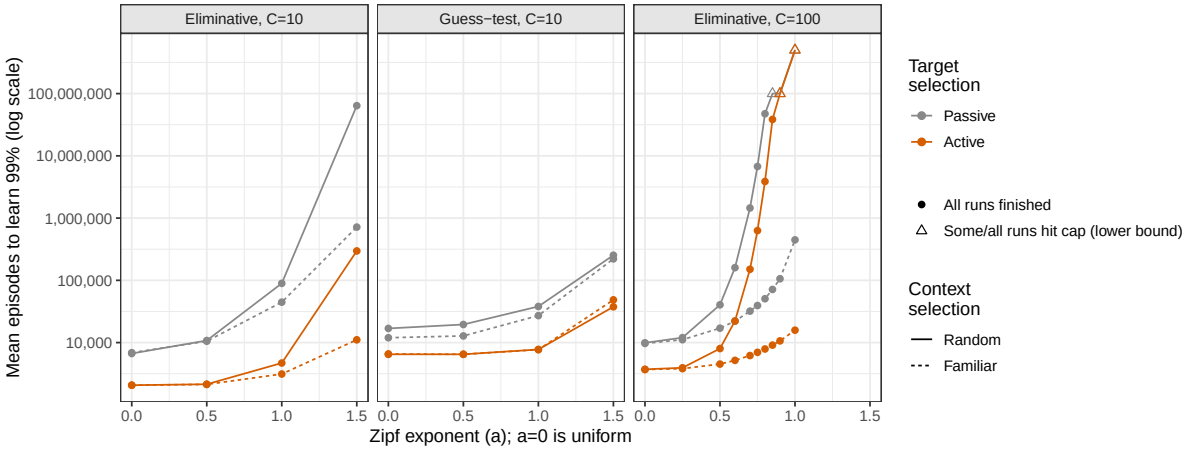


Figure 2. Mean episodes (log scale) to acquire 99% of a 1000-word vocabulary as a function of the Zipf exponent a, for the eliminative and guess-test learners at $C = 10$ and the eliminative learner at $C = 100$, by target selection (color) and context selection (line type). Open triangles mark cells in which some or all runs reached the episode cap (5×10^7 for $a \leq 0.75$, 10^8 for $0.8 \leq a \leq 0.9$, 5×10^8 for $a = 1$); these points are lower bounds.

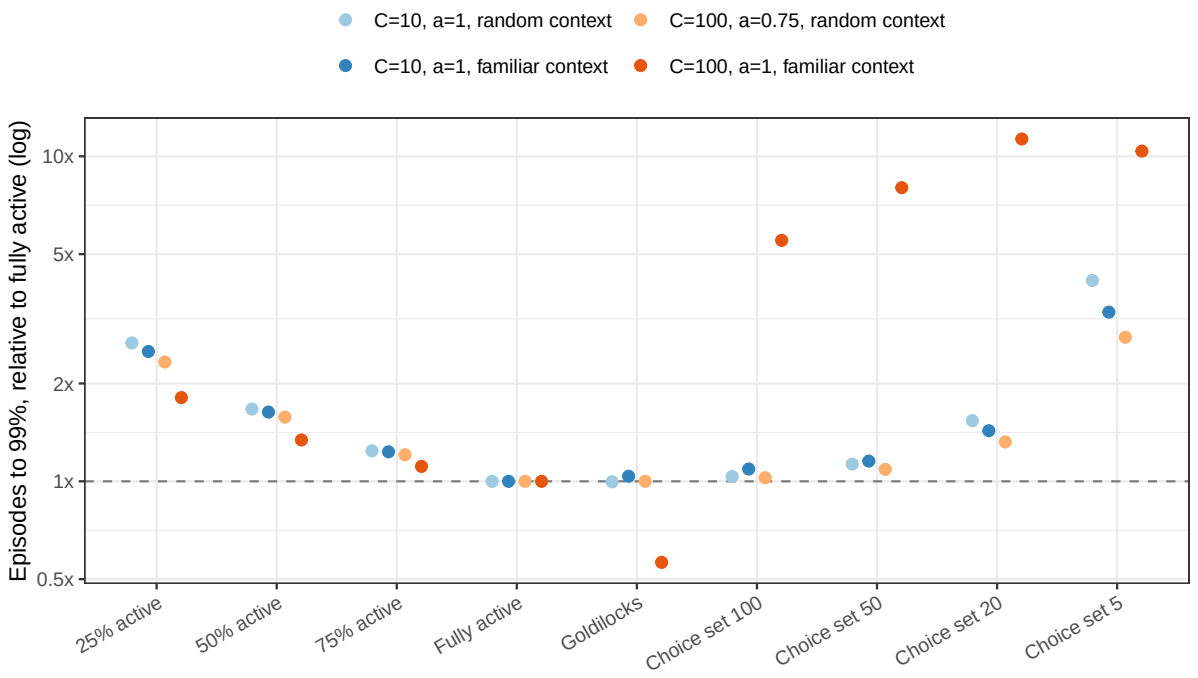


Figure 3. Episodes to learn 99% of the vocabulary relative to fully active selection (log scale; dashed line = fully active), eliminative learner, for partial autonomy, Goldilocks targeting, and bounded choice sets, in four settings. Pure passive sampling (10.6–26.6x) is omitted for scale.

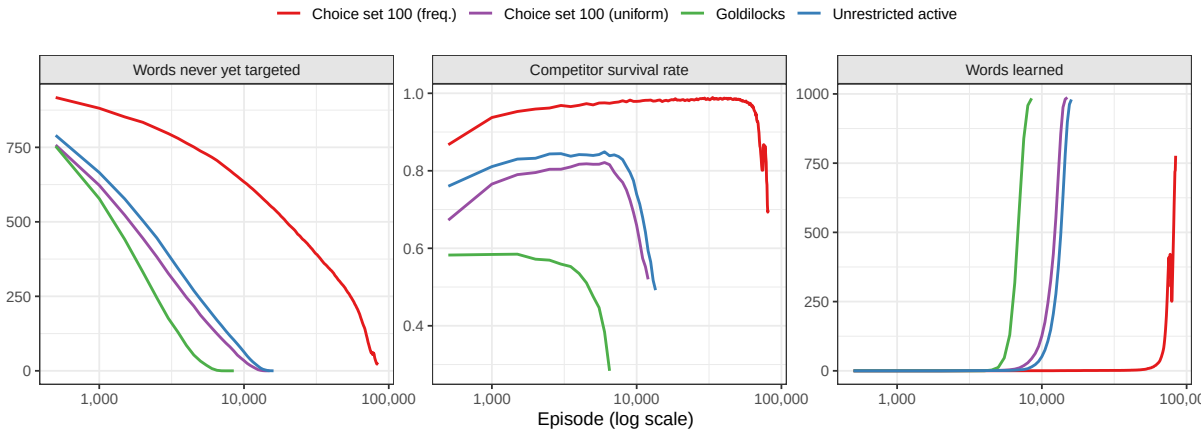


Figure 4. Trajectories under the original familiarity measure, $C = 100$, $a = 1$, familiar context (means of 10 runs). Left: words never yet targeted. Middle: competitor survival, the share of the target's remaining competing referents that reappear among its distractors and so escape elimination, shown until half the vocabulary is learned. Right: words learned. The frequency-weighted choice set leaves many words unencountered for longer, and while it does, familiar-context selection barely reduces competitor survival.

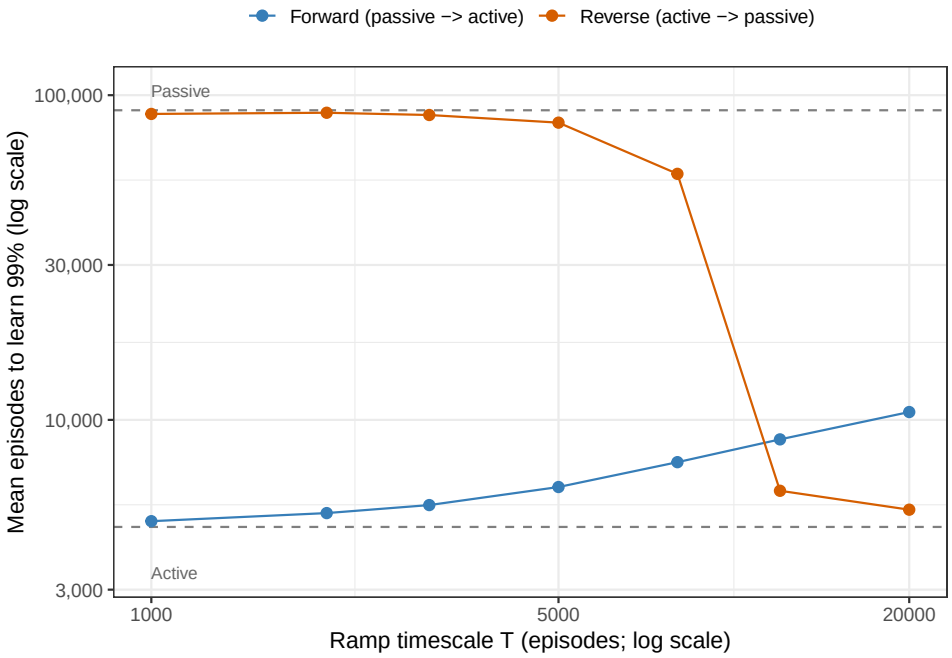


Figure 5. Mean episodes to learn 99% of the vocabulary vs. ramp timescale T, forward (passive-to-active) vs. reverse (active-to-passive) developmental trajectories. Dashed lines mark pure passive and pure active selection (Table 6).

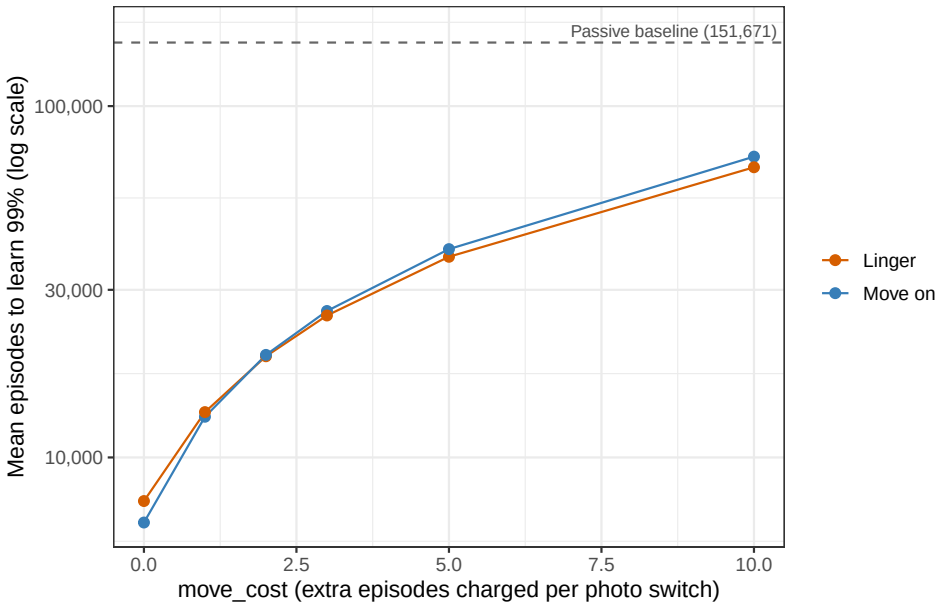


Figure 6. Mean episodes to learn 99% of the vocabulary vs. `move_cost`, active eliminative learner on Flickr8k captions. Moving on to a new photo wins when switching is cheap; lingering on the current photo overtakes it once switching costs more, with the crossover between `move_cost`=1 and 2.